

MA 2320 Final Exam Sample 2

Name: _____

No calculators, no phones, no electronics allowed. Must show work for credit.

1. Compute, using the definition, the exact value of

$$\int_0^4 (3x - x^2) dx$$

2. Compute the area between the graphs of $y = x - 2$, and $x = y^2$.
3. Compute the volume of revolution when the region between the graphs of $y = x^3$, and $y = 4x$ is revolved about the x-axis using either discs or shells.
4. Given $a(t) = 3e^{2t}$, $s_0 = 10$ and $v_0 = 0$. Find $s(t)$ and $v(t)$ Use your results to answer the following

- (a) When is the velocity 2?
- (b) What is the position of the particle after 10 seconds?

5. $\int \frac{e^{4x}}{3e^{4x}+4} dx$

6. $\int x^2 \sin(3x) dx$

7. $\int \sin^3(4x) dx$

8. $\int \frac{1}{\sqrt{x^2-1}} dx$

9. $\int \frac{x^2+x+2}{(x+1)(x^2+1)} dx$

10. For the following determine if the series converges or diverges. For all of these, as in class, show the following: 1. test/series type, 2. Conclusion 3. Criteria 4. work.

(a) $\sum_{n=-3}^{\infty} \frac{4}{3^n}$

(b) $\sum_{n=1}^{\infty} \frac{2n^2 + 1}{3n^3 + 4}$

$$(c) \sum_{n=1}^{\infty} \frac{2^n}{n!}$$

$$(d) \sum_{n=1}^{\infty} \left[1 + \frac{1}{n+1} \right]^n$$

11. Find the interval of convergence and the radius of convergence for

$$\sum_{n=1}^{\infty} \frac{2^n(x-2)^n}{n}$$

12. Find the Taylor/Maclaurin series from the definition for $f(x) = \ln(1+2x)$.
13. Find the Taylor/Maclaurin series from a known series for $f(x) = \ln(1+2x)$.
14. Graph $x^2 + y^2/4 = 1$.
15. Graph $r(\theta) = \cos(2\theta)$.
16. Translate from a parametric equation into a rectangular equation.
- (a) $x = 2t + 1$ $y = 3 - t$.
- (b) $x = t^2$ $y = 1 + t$.
- (c) $x = 2 \cos(t)$ $y = 3 \sin(t)$.